

6032B Energy Conservation & Management

POLY PMNA - Polytechnic Study Hub

Semester 6 | Program Elective | Credits 4 | Periods 60

Downloadable handbook generated from lessons-6032B.html

Core areas: renewable energy sources, energy management, conservation in buildings, HVAC, lighting, BEE star labelling, energy audit and Sankey diagrams.

Course Overview and Roadmap

Objectives: address energy crisis, identify need of energy conservation, observe standard practices based on Energy Conservation Act, and apply good engineering practices to reduce energy loss.

CO1: Explain renewable energy sources and importance. CO2: Summarize energy management techniques. CO3: Explain conservation practices in buildings. CO4: Summarize energy audit in electrical systems.

Roadmap: M1 renewable energy sources. M2 energy scenario, energy management and BEE labelling. M3 energy conservation in buildings, HVAC and lighting. M4 energy audit, instruments, methodology and Sankey diagram.

Module Chapters

Module 1: Energy sources are classified as conventional, non-conventional, renewable, non-renewable, commercial and non-commercial. Solar energy includes PV and thermal conversion. Wind energy systems use rotor, blades, generator, tower and controller. Ocean energy includes tidal and wave energy. Geothermal, biogas and biomass are also covered.

Module 2: Energy scenario includes global reserves, Indian sector-wise consumption, long-term energy scenario and environmental impacts. Energy management means planned measurement, analysis and improvement of energy use. BEE star labelling helps consumers compare appliance energy performance.

Module 3: Building energy conservation includes domestic and commercial saving tips, HVAC components, thermal insulation, lighting energy saving opportunities, LED retrofit, daylight use, occupancy sensors and maintenance.

Module 4: Energy audit includes objectives, preliminary/targeted/detailed audit, instruments, data sheets, phases, ten-step methodology, report format and Sankey energy flow diagram.

Formula Bank, Practice and Answers

Formula bank: Energy = Power x time. Cost = kWh x tariff. Energy saving = old consumption - new consumption. Payback = investment / annual saving. Efficiency = useful output / input x 100 percent.

Solved example: 80 lamps of 40 W replaced by 18 W LEDs. Usage = 10 h/day. Power saved = $80 \times 22 \text{ W} = 1.76 \text{ kW}$. Daily energy saving = $1.76 \times 10 = 17.6 \text{ kWh}$. At Rs.8/kWh, daily cost saving = Rs.140.8.

Expected questions: classify energy sources, explain wind and biogas systems, explain energy management, BEE star labelling, HVAC and lighting conservation, audit instruments, audit methodology and Sankey diagram.

Model answer guide: write measured data, formulas, units, saving, cost saving, payback, diagram points and common mistakes. Do not recommend action without measurement.